

Exercise Sheet 1

Question 1. Which of the following are *not* density matrices?

$$A = \frac{1}{4} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{C}^{2 \times 2}, \quad B = \frac{1}{2} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \in \mathbb{C}^{2 \times 2},$$
$$C = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \in \mathbb{C}^{4 \times 4}, \quad D = (1) \in \mathbb{C}^{1 \times 1}.$$

Question 2. Which of the following density matrices correspond to *pure* states?

$$A = \frac{1}{4} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{C}^{2 \times 2}, \quad B = \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} \in \mathbb{C}^{2 \times 2},$$
$$C = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \in \mathbb{C}^{4 \times 4}, \quad D = (1) \in \mathbb{C}^{1 \times 1}.$$

Question 3. Which of the following density matrices correspond to *classical* states in the standard basis?

$$A = \frac{1}{4} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{C}^{2 \times 2}, \quad B = \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} \in \mathbb{C}^{2 \times 2},$$
$$C = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \in \mathbb{C}^{4 \times 4}, \quad D = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \mathbb{C}^{4 \times 4}.$$

Question 4. Recall that $\mathcal{S}(\mathcal{H})$ denotes the set of density matrices on $\mathcal{H}$. For arbitrary linear operators ρ_1, ρ_2 on $\mathcal{H}$, consider the statements:

(I) $\rho_1 \in \mathcal{S}(\mathcal{H})$ and $\rho_2 \in \mathcal{S}(\mathcal{H})$.

(II) $\rho_1 \rho_2 \in \mathcal{S}(\mathcal{H})$.

Which of the following holds in general?

(A) (I) $\implies$ (II).

(B) (I) $\impliedby$ (II).

(C) (I) $\iff$ (II).

(D) None of those.

Question 5. Consider the pure qubit state

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle.$$

- (a) Write its density matrix $\rho = |\psi\rangle\langle\psi|$ in the standard basis.
- (b) Find the outcome probabilities when measuring in the basis $\{|0\rangle, |1\rangle\}$.
- (c) Find the outcome probabilities when measuring in the basis $\{|+\rangle, |-\rangle\}$, where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$.

Question 6. Which of the following pairs $(\mu(0), \mu(1))$ define a POVM on $\mathbb{C}^2$? Which of the valid POVMs are projective? Here I is the 2×2 identity matrix.

- (A) $(|0\rangle\langle 0|, |1\rangle\langle 1|)$, (B) $\left(\frac{1}{2}I, \frac{1}{2}I\right)$,
- (C) $\left(\frac{1}{2}I, \frac{1}{4}I\right)$, (D) $\left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}\right)$.

For each valid POVM, compute the outcome probabilities for $\rho = \frac{1}{4} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$.

Question 7. Which of the following vectors are Bloch vectors of qubit states? Which of the valid vectors describe pure states, and which describe mixed states?

$$r_A = (0, 0, 1), \quad r_B = \left(\frac{1}{2}, \frac{1}{2}, 0\right), \quad r_C = (1, 1, 0), \quad r_D = (0, 0, 0).$$

For each valid vector, find the probability of outcome 0 in a Z -basis measurement and the probability of outcome + in an X -basis measurement.

Question 8. Let

$$\rho = \frac{1}{4} \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}, \quad O = 2|0\rangle\langle 0| - |1\rangle\langle 1|.$$

- (a) What are the possible outcomes when measuring the observable O , and what are their probabilities in the state ρ ?
- (b) Compute $\mathbb{E}_\rho[O]$ both as a weighted average of the outcomes and using $\text{tr}[\rho O]$.
- (c) Can a single measurement of O return the value $\mathbb{E}_\rho[O]$? Explain what this expectation value means.